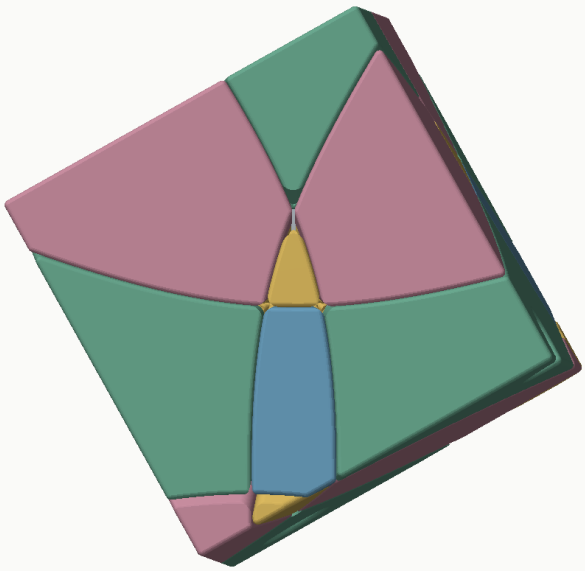
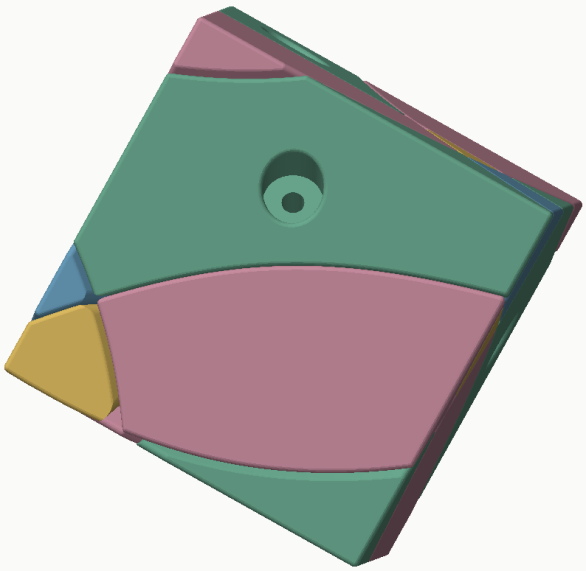
Solve the wonky triangular prism

For the successfully printed 64 mm, non-wavy shape mod. Restore the cube shape, including the orientation of every screwed center.



Front and above



Back and below

The colors only distinguish piece families in this guide. Your puzzle can be one color: use its shape and the solved reference views to recognize each piece's home.

- **5 screwed centers**
- **3 vertical petals**
- **6 horizontal petals**
- **6 corners**

The solving order

1 **Vertical petals** — Place all three, then fix any pair of flips. p. 4

2 **Screwed centers** — Fix the side centers, then the two poles. p. 4

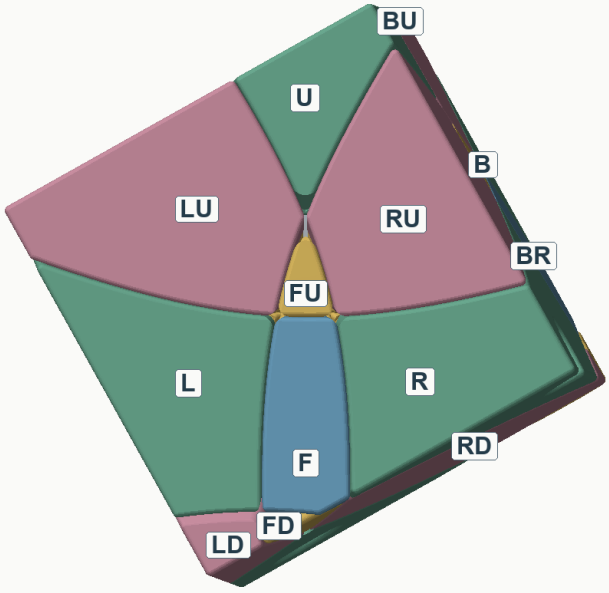
3 **Horizontal petals** — Use three-cycles; their orientation follows their position. p. 5-6

4 **Floating corners** — Finish with corner-only three-cycles. p. 7-8

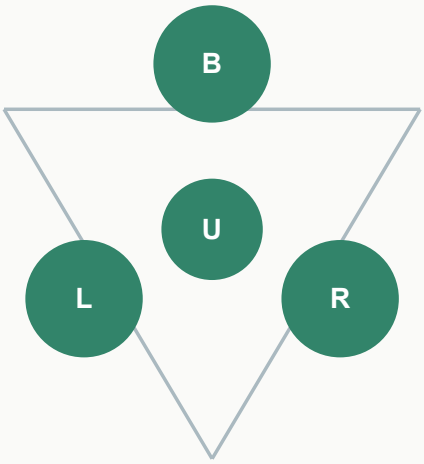
Already solved the petals and all five centers? Go straight to page 7. For the full method, first establish the fixed axis names on page 2 and the slot names on page 3.

Five axes, two move sizes

The prism is tilted inside the cube. U and D are the opposing polar screws, not necessarily the top and bottom faces of the solved cube.



Looking down the U axis



R is the right-front side axis; L is the left-front one; B is behind. D is opposite U. Keep these names fixed even when you change your grip.

Move	Printed center	Amount
R	C02	180 degrees
L	C01	180 degrees
B	C03	180 degrees
U / U'	C04	120 degrees CW / CCW
D / D'	C05	120 degrees CW / CCW

Clockwise means looking from outside straight into that axis's screw hole. Judge D from underneath. R, L and B are half-turns; either direction has the same final effect. U' and D' undo U and D.

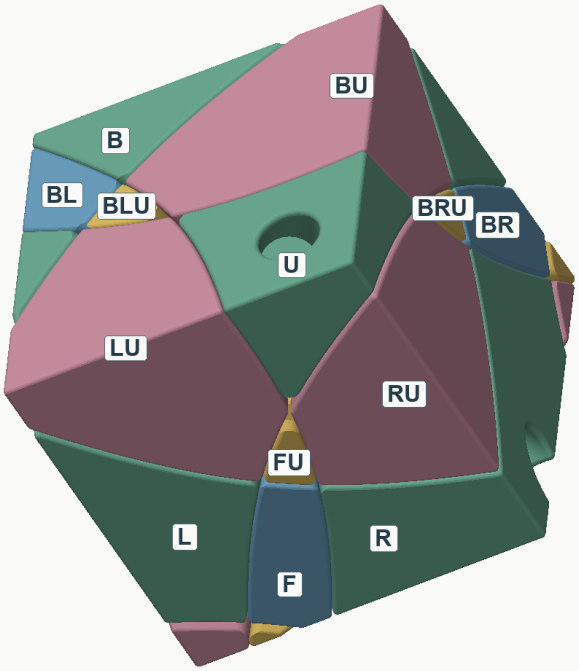
R U' R U **Read moves from left to right.**

Half-turn R; counterclockwise U; half-turn R; clockwise U.

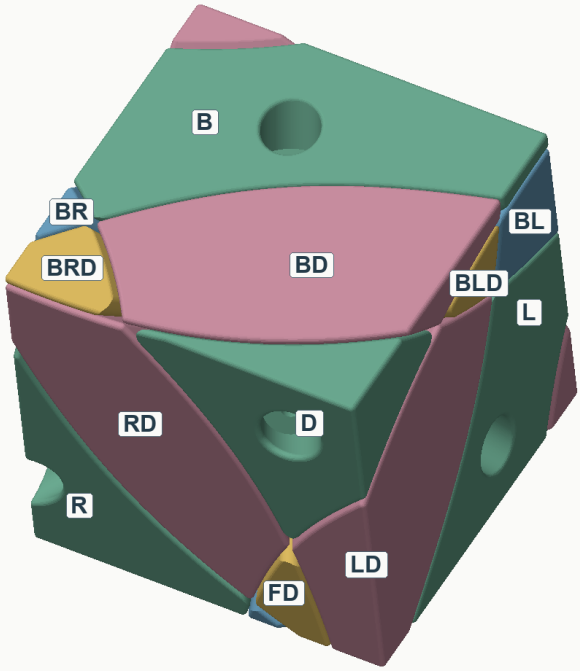
Turn the entire layer attached to the chosen axis, with every piece it carries. Finish each 180- or 120-degree step before starting the next move. The solved cube outline may disappear during an algorithm.

Know which piece belongs where

Labels name fixed slots, not pieces traveling with a turn. These are solved views: use the outlines to identify the correct piece and its final orientation.



Looking toward U



Looking toward D from the back

Three vertical petals

Slot	Location	Printed part
F	Between L and R	V01
BR	Between B and R	V03
BL	Between B and L	V02

Six horizontal petals

Slot	Printed part
RU	H03
LU	H01
BU	H05
RD	H04
LD	H02
BD	H06

Six floating corners

Slot	Printed part
FU	K01
BRU	K05
BLU	K03
FD	K02
BRD	K06
BLD	K04

RU is between the R and U centers. **FU** is the front upper corner between L, R and U; **BRU** is between B, R and U. Replace U with D for the lower layer. There is only one front corner per layer: **FU and FD**.

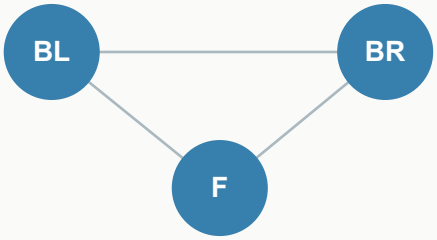
Petals first, then the fixed centers

Work on the three blue vertical petals first. Ignore the horizontal petals and corners until later; they are allowed to move throughout this page.

1a. Put the vertical petals in their home slots

Ignore their orientation for now. Each side half-turn swaps one pair. Choose a swap that puts a misplaced petal into its own slot; repeat. At most two swaps are needed.

Swap these slots	Do
F and BR	**R**
F and BL	**L**
BR and BL	**B**



1b. Flip any two wrongly oriented vertical petals

Once all three are in place, either none or exactly two need a half-turn in their own slots. Use the matching sequence below; it returns all three to the same slots.

Flip this pair	Sequence
F and BR	**L B L R**
F and BL	**R B R L**
BR and BL	**R L R B**

2a. Orient the three side centers

With the vertical petals fully solved, either zero or two side centers are half a turn wrong. Repeat the two named side moves three times. The vertical petals return fully solved.

Wrong side centers	Sequence
R and L	**R L R L R L**
L and B	**L B L B L B**
B and R	**B R B R B R**

2b. Orient U and D directly

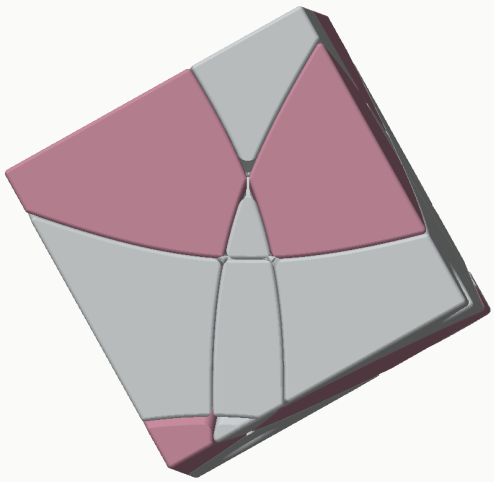
Use U or U' to finish the upper screwed center; use D or D' to finish the lower one. These moves preserve the solved vertical petals and side centers.

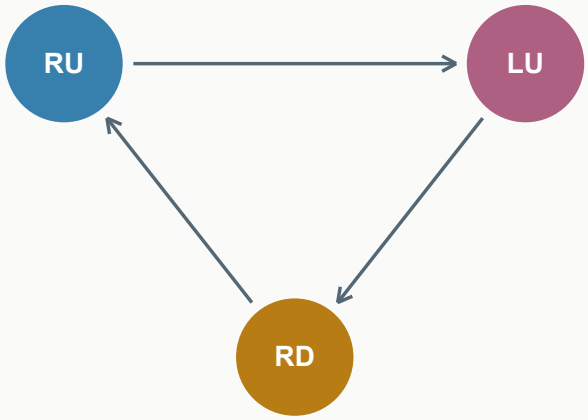
CHECKPOINT: all five centers and all three vertical petals are solved.

Solve six petals with a three-cycle

This stage preserves all five screwed centers and all three vertical petals. The floating corners may move; leave them for the last stage.

```
M = R U' R U
M' = U' R U R
```





M sends RU to LU, LU to RD, and RD to RU. Every other horizontal petal returns to its slot. M' runs the same cycle backwards.

Use a setup to reach any three slots

Page 6 lists all 20 triples. Find the three slots you want, then do **Setup S, Use, Undo S**. For the opposite cycle, change M to M' (or M' to M) and keep both setup columns unchanged.

How to choose a useful cycle

Pick a wrong slot **a**. The piece in a belongs in **b**; the piece in b belongs in **c**. If c is not a, perform **a > b > c > a**. This puts the first two pieces home. If c is a, leave that swapped pair for now and try another wrong slot. Repeat until only swapped pairs remain, or all six are solved.

If only two swapped pairs remain

For **a swaps with b** and **c swaps with d**, do these two cycles:
a > b > c > a, then **a > d > c > a**.

Example: RU / LU swapped, and RD / LD swapped

```
M   then   L M L
```

A horizontal petal in its correct slot automatically has its correct orientation. **There is no separate petal-twisting stage.** Stop when all six are flush with the solved centers and vertical petals.

Any horizontal-petal three-cycle

An arrow means where the current occupant moves. Every listed cycle closes back to its first slot. A dash means no setup.

M = R U' R U

M' = U' R U R

Cycle of occupants	Setup S	Use	Undo S
RU > LU > BU	R U'	**M**	U R
RU > LU > RD	-	**M**	-
RU > LU > LD	D	**M**	D'
RU > LU > BD	D'	**M**	D
RU > BU > RD	U	**M'**	U'
RU > BU > LD	U D	**M'**	D' U'
RU > BU > BD	U D'	**M'**	D U'
RU > RD > LD	L	**M'**	L
RU > RD > BD	D' L	**M**	L D
RU > LD > BD	D L	**M'**	L D'
LU > BU > RD	U'	**M**	U
LU > BU > LD	U' D	**M**	D' U
LU > BU > BD	U' D'	**M**	D U
LU > RD > LD	U' L	**M'**	L U
LU > RD > BD	R D'	**M'**	D R
LU > LD > BD	U' D L	**M'**	L D' U
BU > RD > LD	U L	**M'**	L U'
BU > RD > BD	U D' L	**M**	L D U'
BU > LD > BD	U D L	**M'**	L D' U'
RD > LD > BD	R D L	**M'**	L D' R

Example: RU > BU > LD > RU

U D | U' R U R | D' U'

Setup U D; use M'; undo with D' U'.

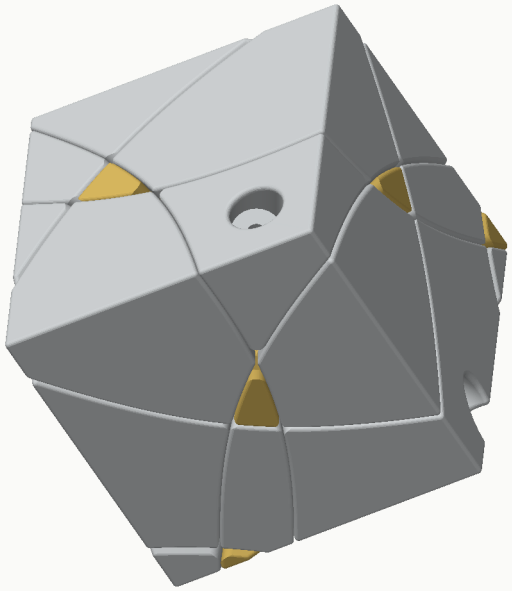
Reverse direction: toggle M / M' only. Always finish the undo column, even if a piece looks solved halfway through. The completed sequence restores every center and vertical petal.

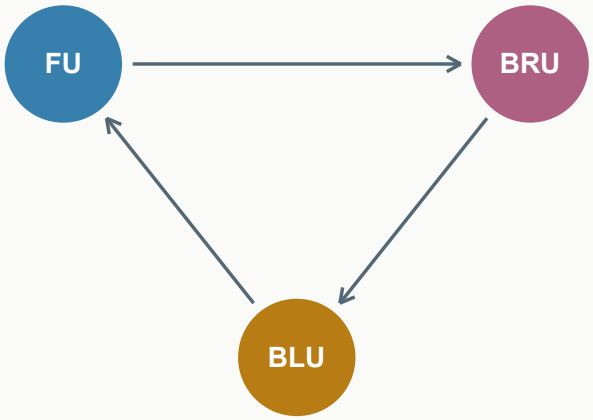
Finish with corner-only cycles

Start here only when all five centers and all nine petals are solved. This ten-move algorithm restores them all, including their orientation.

N = L U' R U R L D R D' R

N' = R D R D' L R U' R U L





N sends FU to BRU, BRU to BLU, and BLU to FU. The three lower corners return unchanged. N' reverses the upper cycle.

Position is orientation

Each corner has only one reachable orientation in a particular slot. Use the solved shapes on page 3 to identify its home; once it is there, it faces the correct way. No final corner-twist algorithm is needed.

Choose a cycle, then use page 8

Follow the same method as with horizontal petals: if the piece in a belongs in b, and the piece in b belongs in c, send **a > b > c > a**. The setup table reaches every triple, including mixed upper and lower corners.

Example: two swapped pairs

If FU / BRU are swapped and FD / BRD are swapped, do **FU > BRU > FD**, then **FU > BRD > FD**.

R U' R | N | R U R

U R | N | R U'

Finish: every piece now belongs in its own slot and orientation, so the outer surface returns to the solved cube. Keep the original axis names through each setup, algorithm and undo.

Any corner three-cycle

Perform Setup S, then the indicated N or N', then Undo S. Only the three named corners change after the entire sequence.

N = L U' R U R L D R D' R

N' = R D R D' L R U' R U L

Cycle of occupants	Setup S	Use	Undo S
FU > BRU > BLU	-	**N**	-
FU > BRU > FD	R U' R	**N**	R U R
FU > BRU > BRD	R U R	**N**	R U' R
FU > BRU > BLD	U R U' B	**N'**	B U R U'
FU > BLU > FD	L U L	**N'**	L U' L
FU > BLU > BRD	U L U' R	**N**	R U L U'
FU > BLU > BLD	R U' B	**N**	B U R
FU > FD > BRD	U R	**N'**	R U'
FU > FD > BLD	U' L	**N**	L U
FU > BRD > BLD	B	**N'**	B
BRU > BLU > FD	U R U' R	**N**	R U R U'
BRU > BLU > BRD	L U' R	**N'**	R U L
BRU > BLU > BLD	R U L	**N'**	L U' R
BRU > FD > BRD	U' R	**N'**	R U
BRU > FD > BLD	L	**N**	L
BRU > BRD > BLD	U B	**N'**	B U'
BLU > FD > BRD	R	**N'**	R
BLU > FD > BLD	U L	**N**	L U'
BLU > BRD > BLD	U' B	**N'**	B U
FD > BRD > BLD	R U R U' B	**N**	B U R U' R

Example: BLU > FD > BRD > BLU

R | R D R D' L R U' R U L | R

Setup R; use N'; undo R. Adjacent R R may be omitted.

Reverse direction: toggle N / N' only. The table is exhaustive for any three corners. Repeat cycles until all six corners match the solved outline.

Checkpoints and troubleshooting

The method solves shape and orientation, not just which family occupies each gap. Use the fixed slot map throughout.

After this stage	What must be fully solved
Vertical petals	F, BR, BL - both position and orientation
Centers	R, L, B, U, D - all five outlines aligned
Horizontal petals	RU, LU, BU, RD, LD, BD
Corners	FU, BRU, BLU, FD, BRD, BLD

The two main algorithms

M = R U' R U

N = L U' R U R L D R D' R

For other slots: Setup, algorithm, undo setup. See pages 6 and 8.

Something solved is now wrong.

Check that you completed the undo setup and kept the same axis names. U and D are 120-degree moves; the three side moves are 180 degrees.

A lower-layer move behaves backwards.

D is clockwise when viewed from below, looking into its screw. A view from above reverses the apparent direction.

Only one vertical petal seems flipped.

With all three vertical petals in their correct slots, flips occur in pairs. Recheck the part identities and which end of each petal belongs up.

Only two pieces seem swapped.

After the earlier checkpoints, a lone swap of horizontal petals is unreachable. A lone corner swap is also unreachable. Recheck identities and earlier stages; two swapped pairs are handled on pages 5 and 7.

A corner seems to need a twist in place.

There is no independent corner twist in this mechanism. Recheck its home slot against the solved atlas. The same applies to horizontal petals.

Verification. The guide's sequences were checked against the design's exact oriented move model: all 96 side-center/vertical-petal states; all 360 permutations in each final phase; and 1,000 random 80-move scrambles solved completely. All 40 setup-table entries were independently checked with the CAD axes and 3D rotation matrices.